Foundations of Linear regression

[Linear regression]

$$\hat{y} = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

1.0 Content Block

when modeling positive quantities (e.g. prices or populations) that vary over a large scale—which are better described using a skewed distribution such as the log-normal distribution or Poisson distribution (although GLMs are not used for log-normal data, instead the response variable is simply transformed using the logarithm function); when modeling categorical data, such as the choice of a given candidate in an election (which is better described using a Bernoulli distribution/binomial distribution for binary choices, or a categorical distribution/multinomial distribution for multi-way choices), where there are a fixed number of choices that cannot be meaningfully ordered; when modeling ordinal data, e.g. ratings on a scale from 0 to 5, where the different outcomes can be ordered but where the quantity itself may not have any absolute meaning (e.g. a rating of 4 may not be "twice as good" in any objective sense as a rating of 2, but simply indicates that it is better than 2 or 3 but not as good as 5). Generalized linear models allow for an arbitrary link function, g, that relates the mean of the response variable(s) to the predictors: E (Y) = g − 1 (X B) {\displaystyle E(Y)=g^{-1}(XB)} . The link function is often related to the distribution of the response, and in particular it typically has the effect of transforming between the (− ∞ , ∞) {\displaystyle (-\infty ,\infty)} range of the linear predictor and the range of the response variable. Some common examples of GLMs are:

2.0 Content Block

− 2 X T Y + 2 X T X β → = 0 ⇒ X T X β → = X T Y ⇒ β ^ → = (X T X) − 1 X T Y {\displaystyle {\begin{aligned}-2X^{\textsf {T}}Y+2X^{\textsf {T}}X{\vec {\beta }}&=0\\\Rightarrow X^{\textsf {T}}X{\vec {\beta }}&=X^{\textsf {T}}Y\\\Rightarrow {\vec {\hat {\beta }}}&=\left(X^{\textsf {T}}X\right)^{-1}X^{\textsf {T}}Y\end{aligned}}}

3.0 Content Block

Finance The capital asset pricing model uses linear regression as well as the concept of beta for analyzing and quantifying the systematic risk of an investment. This comes directly from the beta coefficient of the linear regression model that relates the return on the investment to the return on all risky assets.

4.0 Content Block

y {\displaystyle \mathbf {y} } is a vector of observed values y i (i = 1 , … , n) {\displaystyle y_{i}\ (i=1,\ldots ,n)} of the variable called the regressand, endogenous variable, response variable, target variable, measured variable, criterion variable, or dependent variable. This variable is also sometimes known as the predicted variable, but this should not be confused with predicted values, which are denoted y ^ {\displaystyle {\hat {y}}} . The decision as to which variable in a data set is modeled as the dependent variable and which are modeled as the independent variables may be based on a presumption that the value of one of the variables is caused by, or directly influenced by the other variables. Alternatively, there may be an operational reason to model one of the variables in terms of the others, in which case there need be no presumption of causality.

5.0 Content Block

for each observation i = 1 , … , n {\textstyle i=1,\ldots ,n} . In the formula above we consider n observations of one dependent variable and p independent variables. Thus, Yi is the ith observation of the dependent variable, Xij is ith observation of the jth independent variable, j = 1, 2, ..., p. The values βj represent parameters to be estimated, and εi is the ith independent identically distributed normal error. In the more general multivariate linear regression, there is one equation of the above form for each of m > 1 dependent variables that share the same set of explanatory variables and hence are estimated simultaneously with each other:

6.0 Content Block

where w = (w 1 , w 2 , … , w q) ■ {\displaystyle \mathbf {w} =(w_{1},w_{2},\dots ,w_{q})^{\intercal }} is a weight vector satisfying ∑ j = 1 q | w j | = 1 {\textstyle \sum _{j=1}^{q}|w_{j}|=1} . Because of the constraint on w j {\displaystyle {w_{j}}} , ξ (w) {\displaystyle \xi (\mathbf {w})} is also referred to as a normalized group effect. A group effect ξ (w) {\displaystyle \xi (\mathbf {w})} has an interpretation as the expected change in y {\displaystyle y} when variables in the group x 1 , x 2 , … , x q {\displaystyle x_{1},x_{2},\dots ,x_{q}} change by the amount w 1 , w 2 , … , w q {\displaystyle w_{1},w_{2},\dots ,w_{q}} , respectively, at the same time with other variables (not in the group) held constant. It generalizes the individual effect of a variable to a group of variables in that (i {\displaystyle i}) if q = 1 {\displaystyle q=1} , then the group effect reduces to an individual effect, and (i i {\displaystyle ii}) if w i = 1 {\displaystyle w_{i}=1} and w j = 0 {\displaystyle w_{j}=0} for j ≠ i {\displaystyle j\neq i} , then the group effect also reduces to an individual effect. A group effect ξ (w) {\displaystyle \xi (\mathbf {w})} is said to be meaningful if the underlying simultaneous changes of the q {\displaystyle q} variables (x 1 , x 2 , … , x q) ■ {\displaystyle (x_{1},x_{2},\dots ,x_{q})^{\intercal }} is probable. Group effects provide a means to study the collective impact of strongly correlated predictor variables in linear regression models. Individual effects of such variables are not well-defined as their parameters do not have good interpretations. Furthermore, when the sample size is not large, none of their parameters can be accurately estimated by the least squares regression due to the multicollinearity problem. Nevertheless, there are meaningful group effects that have good interpretations and can be accurately estimated by the least squares regression. A simple way to identify these meaningful group effects is to use an all positive correlations (APC) arrangement of the strongly correlated variables under which pairwise correlations among these variables are all positive, and standardize all p {\displaystyle p} predictor variables in the model so that they all have mean zero and length one. To illustrate this, suppose that { x 1 , x 2 , … , x q } {\displaystyle \{x_{1},x_{2},\dots ,x_{q}\}} is a group of strongly correlated variables in an APC arrangement and that they are not strongly correlated with predictor variables outside the group. Let y ′ {\displaystyle y'} be the centred y {\displaystyle y} and x j ′ {\displaystyle x_{j}'} be the standardized x j {\displaystyle x_{j}} . Then, the standardized linear regression model is